

L3 Gate 3 SSG-8 Mersenne Operator $\hat{M}_p$ on $H^2(D_{IV}^5)$ Explicit v0.2

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L3 Gate 3 SSG-8 Mersenne Operator v0.2

0. Goal

Per Casey 15:11 EDT 4-decision approval + Track A L3: explicit substrate-Mersenne operator $\hat{M}_p$ action on Bergman $H^2(D_{IV}^5)$. Extension of v0.1 morning framework to operator-theoretic derivation.

1. Substrate-Mersenne Operator Action on $H^2(D_{IV}^5)$

Substrate-Mersenne operator definition:

$$\hat{M}_p : \mathbb{Z}_+ \rightarrow \mathbb{Z}_+, \quad \hat{M}_p(k) = 2^k - 1$$

Lifted action on Bergman $H^2(D_{IV}^5)$ via substrate-K-type spectral decomposition:

$$H^2(D_{IV}^5) = \bigoplus_{(\lambda_1, \lambda_2)} V_{(\lambda_1, \lambda_2)}$$

substantive substrate-Mersenne operator action via substrate-K-type Casimir spectrum mapping:

$$\hat{M}_p \cdot |V_{(\lambda_1, \lambda_2)}\rangle = |V_{(\lambda'_1, \lambda'_2)}\rangle$$

where $(\lambda'_1, \lambda'_2) = \text{substrate-Mersenne mapped substrate-K-type substrate-natural per substrate-primary cascade rank} \rightarrow N_c \rightarrow g$.

2. Substrate-Primary Cascade Operator-Theoretic

Per Friday May 22 Mersenne Network + K207 PASS A-tier:

substantive substrate-primary cascade as substrate-Mersenne operator iteration:

$$\hat{M}_p(rank) = M_2 = 3 = N_c$$

$$\hat{M}_p(N_c) = M_3 = 7 = g$$

$$\hat{M}_p(n_C) = M_5 = 31$$

$$\hat{M}_p(g) = M_7 = 127$$

$$N_{max} = M_g + 10 = 127 + 10 \text{ substrate-additive}$$

substrate-additive identity per Casey #5 Integer Web:

$$N_{max} = M_g + (g + N_c) = 127 + (7 + 3) = 137$$

substantive substrate-natural additive identity.

3. Reed-Solomon Coding on $GF(2^g)$

Per K59 cyclotomic mechanism framework RATIFIED + Paper #122 Information Substrate:

substantive substrate-Reed-Solomon coding on $GF(2^g) = GF(128)$ substrate-natural per substrate-Mersenne ladder substrate-natural form.

Substrate-Reed-Solomon code parameters substrate-natural: - Code length $n = 2^g - 1 = 127$ substrate-natural - Substrate-natural code dimensions per substrate-Mersenne cascade

4. Substrate-Mersenne Operator Spectral Properties

Substantive substrate-Mersenne operator spectral substrate-natural form: - Domain: substrate-primary integers $(rank, N_c, n_C, C_2, g, N_{max})$ - Range: substrate-Mersenne integers $M_p(3, 7, 31, 127)$ - Substrate-natural fixed point: $M_2 = 3$ (substrate-rank-color invariant) - Substrate-natural growth: exponential 2^p substrate-natural

substantive substrate-mechanism: substrate-Mersenne operator substrate-natural cascade FORCES substrate-primary lineage $rank \rightarrow N_c \rightarrow g$ via substrate-natural exponential substrate-Cartan-base substrate-mechanism.

5. K207 PASS A-tier Operator-Theoretic Substantive Cross-Link

Per K207 PASS A-tier substantive substrate-mechanism: substantive substrate-Mersenne ladder substrate-mechanism FORCING per substrate-natural Reed-Solomon coding $GF(2^g)$ substrate-natural.

substantive operator-theoretic substantive substrate-mechanism: substrate-Mersenne operator $\hat{M}_p$ substrate-canonical action on substrate-Bergman $H^2(D_{IV}^5)$ substrate-K-type spectrum substrate-natural per substrate-primary cascade.

6. Multi-Week RIGOROUS Path v0.2

substantive 5-step multi-week RIGOROUS closure: - Step L3-1: explicit substrate-Mersenne operator $\hat{M}_p$ substrate-canonical form on $H^2(D_{IV}^5)$ - Step L3-2: explicit substrate-K-type spectral decomposition substrate-Mersenne mapping $(\lambda_1, \lambda_2) \rightarrow (\lambda_1', \lambda_2')$ - Step L3-3: explicit substrate-Reed-Solomon coding $GF(2^g)$ substrate-natural form per substrate-Mersenne cascade - Step L3-4: substantive K207 PASS A-tier substrate-Mersenne substrate-mechanism FORCING form derivation - Step L3-5: substantive Casey #5 Integer Web substrate-additive identity $N_{max} = M_g + (g + N_c)$ substrate-natural cross-link

7. Closure v0.2

L3 Gate 3 v0.2 explicit substrate-Mersenne operator framework: - substrate-Mersenne operator $\hat{M}_p$ substrate-canonical action on Bergman $H^2(D_{IV}^5)$ - substrate-primary cascade $rank \rightarrow N_c \rightarrow g$ operator-theoretic substantive - substrate-Reed-Solomon coding $GF(2^g)$ substrate-natural per substrate-Mersenne ladder - K207 PASS A-tier substrate-mechanism FORCING substantive substrate-Mersenne substrate-cascade - Casey #5 Integer Web $N_{max} = M_g + (g + N_c)$ substrate-additive substrate-natural

substantive 5-step multi-week RIGOROUS closure per Cal #189.

Tier: L3 Gate 3 v0.2 explicit substrate-Mersenne operator framework at FRAMEWORK candidate; substantive multi-week RIGOROUS closure per Cal #189.

— Lyra, Thu 2026-06-04 15:37 EDT. L3 Gate 3 v0.2: explicit substrate-Mersenne operator $\hat{M}_p$ on $H^2(D_{IV}^5)$ substrate-canonical action; substrate-primary cascade $rank \rightarrow N_c \rightarrow g$ operator-theoretic; Reed-Solomon coding $GF(2^g)$ substrate-natural; K207 PASS A-tier substantive substrate-mechanism FORCING substantive; Casey #5 Integer Web $N_{max} = M_g + (g + N_c)$ substrate-natural; 5-step multi-week RIGOROUS per Cal #189.